



COMP30026

Models of Computation

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Lecture 7

Predicate Logic: Semantics



Meaning of a Formula

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3. **True** if discussing $\mathbb{R}$ and $<$ is “less than”.
4. Unconditionally **true** if universe is $\{0\}$.



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Depends on how we interpret it.

1. **True** if trees are non-empty.
2. **False** if trees can be empty.



Valid, Unsatisfiable, Contingent: Examples

Valid

$$\forall x P(x) \vee \exists y \neg P(y)$$



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Contingent

$$\forall x P(x)$$



Universe

- The set of things we are talking about.
- Quantifiers range over the universe, U .
- Non-empty



Interpretation Functions

Non-logical symbols: predicate, function and constant symbols.

Interpretation function: maps every non-logical symbol to its interpretation.

Constant symbol $\mapsto$ object in U .

Predicate symbol $\mapsto$ relation on U .

Function symbol $\mapsto$ function from U to U .



Models

Model: universe **together with** an interpretation function.

$\mathbb{Z}$ with usual +
is a model of $\forall x \exists y (x + y = 1)$.

$\mathbb{N}$ with usual +
is **not** a model of $\forall x \exists y (x + y = 1)$.



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Truth of $P(x)$ depends on choice of x .

Variable assignment: a function $\nu : \text{vars} \rightarrow U$.

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Example

Consider a model M with universe $U = \{\text{red}, \text{blue}\}$ and interpretation function I such that $I(P) = \{\text{red}\}$. Let $v : \text{var} \rightarrow U$ be a variable assignment. Then,

- $P(x)$ is true in M under v if $v(x) = \text{red}$.
- $P(x)$ is false in M under v if $v(x) = \text{blue}$.



Terms and variable assignments

A variable assignment is a function $v : var \rightarrow U$.

But, given an interpretation I , it **extends** to terms:

$$\begin{aligned} v(a) &= d \\ v(g(t_1, \dots, t_n)) &= g(v(t_1), \dots, v(t_n)) \end{aligned}$$

where d is the element of U that I assigns to a , and $g : U^n \rightarrow U$ is the function that I assigns to g .

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Consider $t = f(y, g(x, a))$. Assume I assigns to a the value 1, to f the multiplication function, and to g addition. If $v(x) = 2$ and $v(y) = 1$ then $v(t) =$.

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Truth of Formulas Depends on Their Parts

The truth of a **closed** formula depends only on the model.

We want free variables only to define truth of a formula **compositionally**.

Useful notation:

$$v_{x \mapsto d}(y) = \begin{cases} d & \text{if } y = x \\ v(y) & \text{otherwise} \end{cases}$$

Read this as “the map v , updated to map x to d .”

Defining Truth in Predicate Logic

In a model M (with universe U and interpretation function I), under a variable assignment v :

- **Atomic formulas:** $P(t_1, \dots, t_n)$ is true iff $(v(t_1), \dots, v(t_n)) \in I(P)$.
- **Existential quantifier:** $\exists x F$ is true iff there exists at least one $d \in U$ such that F is true in M under $v_{x \mapsto d}$.
- **Universal quantifier:** $\forall x F$ is true iff $\neg \exists x \neg F$ is true.

Definitions for connectives are same as in propositional logic.



Quantifier Order

$\forall x \exists y$ is not the same as $\exists y \forall x$.

Former: each x has a y that satisfies $P(x, y)$;

Latter: there is a y that satisfies $P(x, y)$ for every x .

$\forall x \forall y$ is the same as $\forall y \forall x$.

$\exists x \exists y$ is the same as $\exists y \exists x$.



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F **semantically entails** G iff every model of F is a model of G . We write $F \models G$.

F is **logically equivalent** to G iff $F \models G$ and $G \models F$. We write $F \equiv G$.

Summarising: Satisfiability and Validity

A closed formula F is

- **satisfiable** iff $M \models F$ for some model M ;
- **valid** iff $M \models F$ for every model M ;
- **unsatisfiable** iff $M \not\models F$ for every model M ;
- **non-valid** iff $M \not\models F$ for some model M .
- **contingent** iff it is satisfiable but not valid.

As in the propositional case, we have

- F is valid iff $\neg F$ is unsatisfiable;
- F is non-valid iff $\neg F$ is satisfiable.



Example of Non-Validity

Consider the formula

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For example, consider the model with universe $U = \mathbb{Z}$, and the predicate P meaning “less than”.

Or, let $U = \{0, 1\}$ and let P mean “equals”.

The formula **is** satisfiable, as it is true, for example, in the model where $U = \{0, 1\}$ and P means “less than or equal”.



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If we negate F (and rewrite it) we get

$$(\exists y \forall x P(x, y)) \wedge (\exists x \forall y \neg P(x, y))$$

The right conjunct is made true only if there is some $a \in U$ for which $(a, b) \notin I(P)$ for all $b \in U$.

But the left conjunct requires that $(a, b) \in I(P)$ for at least some b .

Since F 's negation is unsatisfiable, F is valid.



Another Example of Validity

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F is valid no matter what the term t is.

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To see this, again it is easiest to consider

$$\neg F = (\forall x P(x)) \wedge \neg P(t)$$

The term t denotes some element of the universe U , so $\neg F$ cannot be satisfied.